WORKING DRAFT

INTERNATIONAL STANDARD

ISO/WD 34300

Date and time — Codes for calendar systems

Date et heure — Codes pour les systèmes de calendrier

Reference Number
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Foreword

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This document was prepared by Technical Committee ISO/TC 154, *Processes, data elements and documents in commerce, industry and administration*.

This document is amongst a series of International Standards dealing with the conversion of systems of writing produced by Technical Committee ISO/TC 154, *Processes, data elements and documents in commerce, industry and administration*, WG 5 *Date and time*.

This is the first edition of this document.

Introduction

Calendar systems are essential to the tracking of dates and events.

Date and time tracking applications, such as in computing, archiving, communication and bibliography, all depend on the accurate identification of calendar systems for their performance. Moreover, accuracy of date and time tracking is often paramount to important activities from civilian to military use.

This document describes methods to unambiguously define calendar systems, their properties, and provides a mechanism to assign unique identifiers to these data elements.

This document also sets out the necessary procedures to maintain an international registry of calendar systems. == Scope

This document provides a list of codes to represent calendar systems, metadata to support the usage of those codes, and the corresponding maintenance procedures.

The codes were devised for usage of any application requiring the expression of calendar systems in coded form, including for data interchange.

Date and time — Codes for calendar systems

1 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO 639-1, *Codes for the representation of names of languages — Part 1: Alpha-2 code*

ISO 639-2, *Codes for the representation of names of languages — Part 2: Alpha-3 code*

ISO 3166-1, *Codes for the representation of names of countries and their subdivisions — Part 1: Country code*

ISO 5127 (all parts), *Information and documentation – Foundation and vocabulary*

ISO 15924:2004, *Information and documentation — Codes for the representation of names of scripts*

2 Terms and definitions

For the purposes of this document, the terms and definitions given in ISO/CD 34000 and the following apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

— ISO Online browsing platform: available at http://www.iso.org/obp

— IEC Electropedia: available at http://www.electropedia.org

2.1
code
data representation in different forms according to a pre-established set of rules

[SOURCE: ISO 639-2]

2.2
calendar code
code(2.1) used to represent a calendar system

2.3
language code
combination of characters used to represent the name of a language or languages

[SOURCE: ISO 5127, 3.2.5.14]

2.4
script code
combination of characters used to represent the name of a script

[SOURCE: ISO 15924:2004, 3.8]

3 Calendar system models overview

The data models are arranged as shown in the following structure.

```
diagram TopDown {
  title "Calendar system model overview in UML"
```

```
  include ../models/CalendarSystem.lutaml
  include ../models/CalendarEra.lutaml
}

//  relations:
//    - source: CalendarEra
//      target: CalendarSystem
//      direction: 'up'
//  fidelity:
//    hideMembers: true
//    hideOtherClasses: true
```

Figure 1

4 Data types

4.1 Core data types

These are the core data types used within this document.

— `CharacterString`

— `DateTime`, `Date`, `Time`

— `Number`, `Integer`, `Decimal`, `Real`

— `Vector`

— `Boolean`

4.2 Common data types

The following data models are used by other data models specified in this document.

```
diagram DataTypes {
  title "Data types used in this document"

  include ../models/iso639Code.lutaml
  include ../models/iso639Code.lutaml
  include ../models/LocalizedTag.lutaml
  include ../models/LocalizedString.lutaml
  include ../models/JulianDate.lutaml
  include ../models/CalendarDate.lutaml
}
```

Figure 2

5 Data models

```
diagram CalendarSystem {
  title "Calendar system and era models"

  include ../models/CalendarSystem.lutaml
  include ../models/CalendarEra.lutaml
}

// imports:
//   CalendarSystem:
//   CalendarEra:
// groups:
//   # - - AddressProfile
//   # - - FormTemplate
```



```
//   #   - DisplayTemplate
// relations:
//   - source: CalendarEra
//     target: CalendarSystem
//     direction: 'up'
//   # - source: FormTemplate
//   #   target: DisplayTemplate
//   #   direction: '[hidden]right'
```

Figure 3

6 Codes and identifiers

6.1 Requirements

6.1.1 General

A calendar system is eligible for assignment of an entry based on its usage of date identification and the need for international interchange.

Usage of a calendar system is determined by at least one of the following conditions, by presentation of evidence:

— The system has been approved for official use at some level of government (current, historic, or will be in the near future);

— The system has been used to identify with dates, such as in bibliographic information (current, historic, or will be in the near future).

Justification should be given to indicate that international interchange is necessary.

A similar set of requirements apply to the registration of calendar eras. A calendar era entry must be associated with at least one calendar system entry.

6.2 Principles for allocation of code elements

6.2.1 Relationship with names

Calendar systems have codes in the following format:

```
[system-code]
```

Figure 4

EXAMPLE `gregorian` or `gre` for the Gregorian calendar, whose specification is provided in ISO 8601-1.

6.2.2 Construction of the alphabetic code

The following rules are to be adhered to for the assignment of the alphabetic code:

— The alphabetic code uses combinations, in lower case, of between 3 and 20 fixed letters of the 26-character Latin alphabet.

— Codes shall encourage descriptive and distinguishable alphabetic names.

An additional short alias for the alphabetic code composed of 3 fixed letters of the 26-character Latin alphabet is allowed.

6.2.3 User assigned code elements

If users need code elements to represent calendar systems not included in the calendar code registry, the code prefix of `zz` can be used.

The code length for the calendar system identifier must be 3 letters.

Such calendar system identifiers are in the following format:

```
zz[calendar-system-identifier]
```

Figure 5

NOTE Users are advised that the above series of codes are not universals, those code elements are not compatible between different entities.

6.3 List of calendar system and their code elements

The list of items composing the content of the country code is initially compiled in a separate dataset outside of this document. Additional and new entries will be listed by the ISO 34300/AG.

Data attributes provided in the list is defined in Clause 6.

Bibliography

[1] ISO 8601-1, *Date and time — Representations for information interchange — Part 1: Basic rules*

[2] ISO/IEC 10646-2, *Information technology — Universal Multiple-Octet Coded Character Set (UCS) — Part 2: Supplementary Planes*

[3] ISO/IEC TR 15285:1998, *Information technology — An operational model for characters and glyphs*

[4] ISO/CD 34000[1)], *Date and time — Concepts and vocabulary*

[5] The XXIIIrd International Astronomical Union General Assembly, *The XXIIIrd International Astronomical Union General Assembly*, 1997. Available at: https://www.iers.org/IERS/EN/Science/Recommendations/resolutionB1.html

1) Under preparation. (Stage at the time of publication ISO/CD 34000).

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